```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import Lasso
from sklearn import metrics

insurance = pd.read_csv('/content/sample_data/insurance.csv')
insurance.head()
insurance.shape
insurance.info
insurance.describe
insurance.isnull().sum()

print(insurance.age.value_counts())
print(insurance.sex.value_counts())
print(insurance.smoker.value_counts())
print(insurance.charges.value_counts())

LE = LabelEncoder()
insurance['sex'] = LE.fit_transform(insurance['sex'])
insurance['smoker'] = LE.fit_transform(insurance['smoker'])
insurance['region'] = LE.fit_transform(insurance['region'])

X = insurance.drop(['sex', 'smoker', 'region'], axis=1)
Y = insurance['charges']
print(X)
print(Y)

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)
```

```
37    25
59    25
39    25
36    25
38    25
62    23
60    23
63    23
61    23
64    22
Name: count, dtype: int64
sex
male      676
```

```
...
1333    10600.54830
1334     2205.98080
1335     1629.83350
1336     2007.94500
1337    29141.36030
Name: charges, Length: 1338, dtype: float64
```

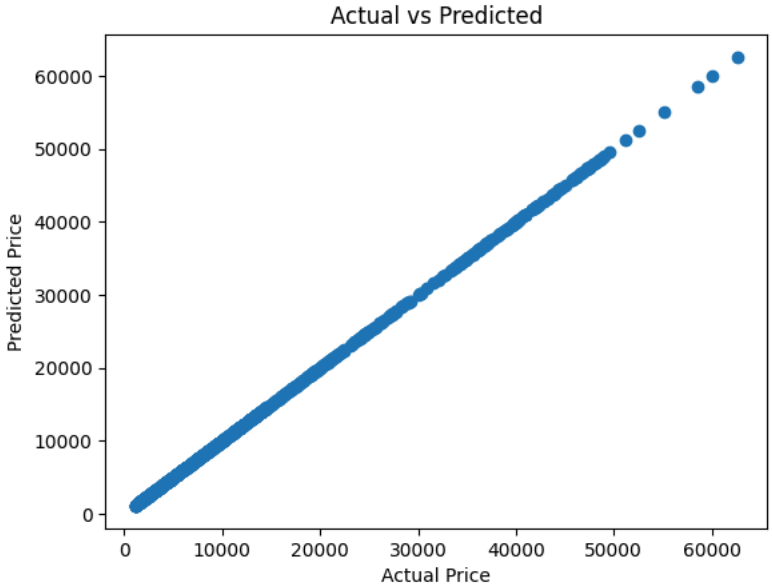
```
lin = LinearRegression()
lin.fit(X_train, Y_train)
Y_pred = lin.predict(X_train)
score = metrics.r2_score(Y_train, Y_pred)
print('R Squared Error:', score)

plt.scatter(Y_train, Y_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted")
plt.show()

Y_test_pred = lin.predict(X_test)
error_score = metrics.r2_score(Y_test, Y_test_pred)
print('R Squared Error:', error_score)

plt.scatter(Y_test, Y_test_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted")
plt.show()
```

```
R Squared Error: 1.0
```




```
R Squared Error: 1.0
```